Ahnaf Shahriar

Email | LinkedIn | Github |

Education

University of Waterloo Waterloo, ON
Bachelor of Applied Science in Computer Engineering *Sept. 2021 – Apr. 2026*

- Recipient of Richard & Elizabeth Madter Entrance Scholarship and President's Scholarship of Distinction
- **Relevant Courses**: Algorithms and Data Structures II, ML for chip Design, Distributed Computing, Compilers, Real-Time Operating Systems

Experience

Undergraduate Research Assistant May. 2025 – Present
University of Waterloo *Waterloo, ON*

- **Research**: Conducting Case study on Static Analysis type checking for Java.
- **Optimization**: Implementing bytecode storage on Nullness checker for EISOP framework.

Wireless Firmware Developer Jan. 2025 – Apr. 2025
ON Semiconductor Corporation *Waterloo, ON*

- **SDK Development**: Integrated *I3C support* and *BLE* profile services into sample code.
- **Wireless Connectivity**: Developed and integrated cutting-edge *LE Audio, HFP, and A2DP* functionality into an all-in-one Application.
- **Wireless Manager**: Modularized and implemented a state machine for power efficiency by *25%*
- **Test Automation**: Automated SDK test cases, reducing manual testing efforts by up to *80%*

IC Design and Verification Intern May 2023 – Aug. 2023, Jan. 2024 – May 2024
NXP Semiconductors Canada *Kanata, ON*

- **IP Design**: Designed multiple IP blocks including *proprietery ECC IP* for dataplane processing chips.
- **Timing Analysis**: Spearheaded critical path improvements in IP block to increase Clock Frequency by *50%*.
- **Functional Testing**: Designed brand new End-to-End functional tests in simulating *10GBps+* traffic for IP.
- **Unit Test Planning**: Created Simulation scenarios for testing High speed Dataplane Processing features.

Embedded Software Engineering Intern Sept. 2022 – Dec. 2022
Synapse Product Development *Seattle, WA*

- **Prototyping**: Leveraged Zephyr RTOS to create a proof of concept on *NRF52 BLE* device.
- **Python APIs**: Developed company specific lab automation software for equipment from *Agilent, Keysight, NI, Tektronik.*
- **Automation**: Streamlined testing and in house procedures using *Python* and *Bash.*
- **Driver Development**: Designed and implemented *drivers* of automated PCB testing Device*(I2C, UART)*

Projects

VHDL Compiler | *Java*

- Performed **Syntactic and Semantic** analysis in object oriented visitor design approach.
- Generated diagrams for circuits in **NAND and CNF** form, with optional **waveform simulation** using bitstreams.
- Implemented modular **x86/ARM ASM and Java** code generation and simulation layers to support cross-platform execution and optimization.

CubeSolver | *C++, Linux, NCurses, OpenGL*

- A Program that can solve **any Rubix Cube** you scramble. Optimized for bitwise operations.
- Implemented optimized **Korf's Algorithm** for solving the cube in **less than 20 moves**.
- Optional 3D Visualization using **OpenGL** for movements.

Technical Skills

Languages: C/C++, Python, Rust, Java, Javascript, SQL, Bash and shell scripting, ASM
Frameworks: LLVM, OpenGL, ROCm, Node Js, ElectronJS, Apache Spark, Hadoop
Developer Tools: Git, Docker, Kafka, Qemu, Matlab, Jupyter Notebook
Libraries: pandas, NumPy, Sci-kit, Matplotlib, pyTorch